

Problem and context

Rapidly onboarding a large construction workforce using shared immersive training

Case summary

- Client: AtkinsRealis | Location: Newfoundland, Canada | Date: 2024
- Solution: 6-metre immersive cylinder with floor projection.
- AtkinsRealis and SDP partners used the Igloo as a training hub for a major offshore construction project.
- The project supported construction of the base of a fixed drilling platform offshore Newfoundland.



Source case image: shared training environment

Problem to solve

- A large workforce needed to be brought up to speed quickly before a complex, time-critical construction phase.
- Training had to communicate site facilities, health and safety protocols, project plans, and working-at-height conditions.
- Screen-based training could not easily create a shared sense of scale, spatial context, or site familiarity.
- The team needed high training turnover without losing confidence, safety focus, or practical readiness.

Why it mattered

- The Concrete Gravity Structure phase involved building a 100-metre conical shaft.
- Overall height increased from 45 metres to 145 metres in under two weeks.
- Any stoppage during this phase could be very costly.
- Staff needed to understand scale, sightlines, spatial dynamics, and site conditions before work began.



Source case image: BIM/site content

Takeaway: AtkinsRealis needed shared immersive training that could turn BIM and site media into workforce readiness at speed.

Igloo solution and results

A 6-metre shared VR cylinder, BIM/Unity workflow, and measurable training throughput

What Igloo delivered

- 6-metre immersive cylinder with floor projection.
- 8 Epson projectors powered the wraparound walls and projected floor.
- Igloo Immersive Media Player II handled playback and room display.
- Floor projection stitched content across walls and floor, essential for conveying working at height.



Example Igloo type: immersive cylinder

Content workflow

- Powered by Igloo Core Engine, the operating system for immersive rooms.
- Igloo Core Engine supported industry-standard training and 3D design content.
- AtkinsRealis used the Unity integration to show realtime interactive content across the display.
- The Creative Design team used Unity and project BIM data to build self-guided virtual walkthroughs.
- 360 images and video from the site were also displayed inside the Igloo.



Source case image: control and content workflow

Results and value

900+

**personnel inducted
in less than two weeks**

15

**people trained
at once**

- Floor projection and True Perspective helped communicate scale, sightlines, and spatial dynamics.
- Shared VR helped staff prepare, reduce risk, and prevent costly mistakes before site work.
- AtkinsRealis plans to expand this model for future large-scale construction projects.

Quote: "The Igloo gives you a true sense of scale."